Lab 02: Evaluate a Detection Rule

Practice cases, confusion-matrix metrics, and transparent reporting

Objective

Evaluate a simple rule against labelled training scenarios. The important result is not a high score by itself; it is understanding what the rule catches, what it misses, and which labels or assumptions affect the result.

USE ONLY TRAINING DATA The bundled cases are synthetic and intended for learning. They are not measured real-world attack performance.

Step 1 — Review the included cases

Open website/public/labs/detection-evaluation-cases.csv, or download the CSV from this lab card. Each row includes a case ID, a labelled scenario, process, command line, and rationale. The companion Python script applies a deliberately narrow sample rule and calculates its results. All cases are synthetic.

From the website folder, run: python public/labs/evaluate-detection-cases.py public/labs/detection-evaluation-cases.csv

Step 2 — Count outcomes

Actual label	Rule alerts	Classification
Suspicious	Yes	True positive (TP)
Benign	Yes	False positive (FP)
Benign	No	True negative (TN)
Suspicious	No	False negative (FN)

Step 3 — Use the metrics

Metric	Formula	What it tells you
Accuracy	(TP + TN) / all cases	Overall correctness; can mislead on imbalanced data.
Precision	TP / (TP + FP)	How many alerts were true positives in the labelled set.
Recall	TP / (TP + FN)	How many labelled suspicious cases were detected.
Specificity	TN / (TN + FP)	How many labelled benign cases were correctly ignored.
F1	2 × P × R / (P + R)	Harmonic mean of precision and recall.

Step 4 — Experiment

1. Record the baseline counts (TP, FP, TN, FN) before changing a rule.
2. Inspect each false positive. Is the rule too broad, or is context missing?
3. Inspect each false negative. What relevant field or event pattern is missing?
4. Change one condition at a time in your own local practice copy.
5. Re-run the same dataset and compare counts. Keep both baseline and revised results.
6. Do not tune on a small set and then present that same score as independent validation.

Step 5 — Write a reproducible report

- Dataset filename and version.
- Number of cases and label distribution.
- Rule logic and required log fields.
- TP, FP, TN, FN and calculated metrics.
- Two examples of false positives or false negatives, if present.
- Limitations, environment, and the next experiment.

ADVANCED HABIT Separate development data from a held-out evaluation set when the dataset is large enough. Review labels, duplicates, and scenario coverage.

A real lab for these notes

Topic: Detection engineering, labelled examples, and performance metrics. These pages connect the existing OpenCyberPath notes to specific provider-hosted exercises. The card label is intentionally generic; the exact task name and direct URL are listed here so you can open the right exercise instead of a whole catalogue.

Concept checkpoints

- Treat the bundled labelled CSV as a fixed, synthetic test set.
- TP = suspicious and alert; FP = benign and alert; TN = benign and no alert; FN = suspicious and no alert.
- Precision is TP/(TP+FP); recall is TP/(TP+FN). Zero denominators need careful handling.
- A tiny synthetic score is a teaching aid, not a real-world performance claim.

Sigma 101 *[CyberDefenders]*
https://cyberdefenders.org/blueteam-ctf-challenges/sigma-101/
Access note: Direct challenge page for a Sigma detection exercise. Sign-in, downloadable artefacts, and availability/access tier can vary; verify the page before starting.

Splunk: Exploring SPL *[TryHackMe]*
https://tryhackme.com/room/splunkexploringspl
Access note: Sign-in required; verify current access. The room provides hands-on SPL practice in its own environment.

ACCESS CHECK Open the exact task page and verify the current login, account tier, VM launch button, free/paid terms, and any cost notes before starting. A provider may change availability or retire a challenge.

Before you click Start

- Write down the provider's target/scope and use only the instance it creates for you.
- Keep this notebook beside the exercise. Use its original course data before trying commands on a local machine.
- Capture timestamps, filenames, row/packet IDs, and a short explanation; redact credentials, tokens, and personal data.
- If a page says retired, premium, or downloadable-only, use the alternative listed in the access note or return to the matching interactive lab above.

Commands you can explain

The evaluator reads fictional CSV rows and prints metrics. Never execute command-line strings from the data.

First download the matching sample file(s) from the existing card when a command names one. Keep the data in a practice folder and never execute strings stored in evidence.

PYTHON

```
python .\evaluate-detection-cases.py .\detection-evaluation-cases.csv
```

Run the supplied local evaluation script using the provided CSV. Keep both files in the same directory.

POWERSHELL / INSPECT LABELS

```
Import-Csv .\detection-evaluation-cases.csv | Group-Object activity_label | Select-Object Name,Count
```

Check the number of benign, suspicious, and ambiguous examples before interpreting results.

POWERSHELL / INSPECT MATCHES

```
Import-Csv .\detection-evaluation-cases.csv | Select-Object
case_id,activity_label,process,command_line
```

Read the inputs and labels side by side; do not execute command_line values.

LINUX / QUICK VIEW

```
cut -d, -f1-4 detection-evaluation-cases.csv | head -n 10
```

Inspect case ID, expected label, process, and command line in the fixture.

PYTHON VERSION CHECK

```
python --version
```

Confirm the interpreter is available before running the local script.

COMMAND HABIT Check the current working directory, shell, target, and expected effect before pressing Enter. Save the exact command and the output that mattered.

Guided lab run

- 1. Download the detection CSV and Python script from the existing lab card.
- 2. Run the script; record TP, FP, TN, FN, ambiguous count, precision, recall, specificity, and F1.
- 3. Open each false positive and false negative; explain the exact rule limitation rather than hiding it.
- 4. Make one rule change at a time, re-run the same labelled cases, and compare the results.
- 5. Write a short evaluation note including limitations, sample size, and the reason for any rule change.

Evidence checklist

- [ ] What exact lab instance or sample file was used?
- [ ] Which command/query and time range produced the observation?
- [ ] What is the directly observed fact, separate from your inference?
- [ ] What benign/alternative explanation could fit?
- [ ] What extra record or control would help confirm the conclusion?

Check your understanding

Q Why exclude ambiguous cases from the binary confusion matrix?

Think it through: Their label intentionally does not settle benign-versus-suspicious status, so forcing a binary answer would misrepresent the test.

Q Can 100% precision on nine toy rows prove production quality?

Think it through: No. It says only how the rule behaved on those particular synthetic rows.

Q What usually changes when a detection is made broader?

Think it through: It may catch more variants (higher recall) but can also alert on more benign cases (more false positives).

NOTEBOOK CLOSE-OUT Finish with: Observation / Evidence source / Why it matters / Alternative explanation / Next check / Limitation. A careful "not enough evidence yet" is a valid professional result.